CNCF ISRAEL MEETUP

Life After NGINX Ingress Controller

Architecting the exit - and a low-downtime live migration

Aviv Kabesa

01 - THE HEADLINE

NGINX Ingress Controller Is Dead, and You Are Probably Running It

24 Mar 2026

The repository was archived and the solution is deprecated. Meaning: no releases, no bug fixes, no CVE patches.

~50%

of Kubernetes clusters still run it as their ingress data plane.

8 CVEs

already accumulated, led by CVE-2026-42945 - a critical heap overflow ("NGINX Rift") rated 9.2 under CVSS v4.0. Permanently unpatched.

SPEAKER

Aviv Kabesa

Cloud Solutions Architect, Microsoft

Previously Viz.AI

DevOps Group Lead at TechGym

Love trekking

02 – DISAMBIGUATION

What Exactly Died, and What Did Not

NGINX Ingress Controller (community)	kubernetes/ingress-nginx – SIG Network	DEPRECATED	free · Apache 2.0 no support to buy
NGINX Ingress Controller	nginx/kubernetes-ingress – F5	ACTIVE	free on NGINX OSS paid on NGINX Plus
NGINX Gateway Fabric	F5 – Gateway API implementation	ACTIVE	free on NGINX OSS paid on NGINX Plus
NGINX, the web server	nginx.org – unrelated release train	UNAFFECTED	free on NGINX OSS paid on NGINX Plus

03 – ROOT CAUSE

An Architecture Problem, Not a Staffing Problem

FAULT 01

Reload storms

Any config change regenerates the whole `nginx.conf` from a Go template and reloads. In-flight connections drop; busy clusters reload constantly.

FAULT 02

Snippet injection

Any namespace could inject raw NGINX config into a shared process through `configuration-snippet` . Disabled by default since v1.9 - still enabled in plenty of clusters. Unmaintainable, and an attack surface.

FAULT 03

Unauthenticated webhook

Validation ran `nginx -t` on untrusted input with no auth. Network reachability equalled code execution - the root of CVE-2025-1974.

04 - RISK

Is This Actually Your Emergency?

Answer two questions before you plan anything

01 Is the ingress internet-facing, or internal to a VPC?

02 Is the admission webhook network-reachable from untrusted pods?

Two NO's might mean you have time.
The EOL is real; the urgency is not evenly distributed.

The toxic combination

Risk is real where exposure + vulnerability + access to sensitive data all meet. Absent that overlap, plan properly instead of rushing.

The log4j lesson

Critical to fix - but not for every service, on day one. Spending all your effort on one loud threat is how the quiet one gets in.

05 – PROPOSAL

Swap First, Gateway API Later

1 NOW

Swap the controller

Same Ingress objects. New binary behind them.

→

2 WHEN READY

Adopt Gateway API

Per route, on your own schedule.

The Shortlist

Traefik

Envoy Gateway

Istio

Cilium Gateway

Cloud controllers

NGINX OSS

Traefik v3.7

both APIs, one instance
free · MIT

Nothing in your cluster changes shape

your Ingress objects		Traefik provider		typed middlewares
unchanged, still class `nginx`	→	reads 80+ annotations at runtime	→	in memory, no config file

0

manifest rewrites to swap

0

reloads - Go router, no NGINX binary

snippets

not translated - reported, not applied

08 - CANDIDATE

Envoy Gateway

Gateway API only free · Apache 2.0 · CNCF

file reload	render config	→	reload process	→	drain connections
xDS	control plane	→	gRPC stream	→	live proxy fleet

provisioning

a `Gateway` spawns its own Envoy Deployment + Service

policy

rate limit, OIDC, JWT, retries - as CRDs

the trade

no Ingress API - every manifest rewritten first

Istio and Cilium

free · vendor / enterprise builds paid

Istio mesh optional

Gateway resource

provisions a dedicated Envoy deployment

ambient (optional)

ztunnel = L4 + mTLS · waypoint = L7 where attached

Cilium built into the CNI

L4

eBPF, in-kernel - no proxy hope

L7

Envoy, per-node DaemonSet - no extra controller

Worth it only if you already run them. Neither reads NGINX annotations.

Cloud Controllers and F5 NGINX

the two people forget

Cloud controllers

cloud LB → pods

data plane lives outside the cluster

Nothing to patch · single-cloud

controller free · LBs billed hourly

F5 NGINX

LB → NGINX pod → pods

different codebase - still a full migration

Keeps annotations + NGINX templating

free on OSS · paid on Plus

06 - LANDSCAPE

Six Options, Six Starting Points

Traefik

both APIs · MIT

free · Traefik Hub is paid

Pure Go, no NGINX binary, no reloads. v3.7 reads 85+ NGINX annotations natively. Lowest-friction swap.

Envoy Gateway

Gateway API only

free · Apache 2.0, CNCF

Reference implementation on Envoy xDS. Declarative rate limiting, OIDC, AI Gateway extensions.

Istio

only if you run the mesh

free · vendor builds paid

Replaces the NGINX Ingress Controller with zero new components. Ambient: ztunnel L4/mTLS, waypoints for L7.

Cilium Gateway

only if it is your CNI

free · Isovalent Ent. paid

Built in when Cilium is installed. eBPF L4 in-kernel, Envoy for L7 in userspace.

Cloud controllers

single-cloud only

controller free, LBs billed hourly

AWS LBC, Azure AGC and GKE Gateway. Real cloud LBs, not in-cluster proxies.

NGINX OSS

F5, Ingress API only

free · NGINX Plus paid

Different codebase - still a migration. Keeps you on annotations, and on NGINX's templating model.

Comparison Matrix

	Traefik v3.7	Envoy Gateway	Istio	Cilium GW	AWS LBC
Ingress API	Yes	No	Yes	Yes	Yes
NGINX annotations	**85+ native**	None	None	None	None
Architecture	Pure Go, dynamic	Envoy / xDS	ztunnel / waypoint	eBPF / Envoy	Cloud LB
Reload problem	None	None (xDS)	None	None (eBPF)	N/A
L7 rate limiting	Middlewares	Yes, token cost	Yes	Basic	Cloud-managed
Complexity	Low	Medium	High	Medium-high	Low
Maturity	Since 2016	Moderate	Very	Moderate	GA 2026
Cost	Free (Hub paid)	Free	Free	Free (Ent. paid)	LB billed hourly
Best for	Drop-in swap	Greenfield GW API	Existing mesh	Existing CNI	Single-cloud

Choosing: First YES Wins

Count Annotations, Not Ingresses

S3	**Snippets, Lua**	weeks to months per snippet cluster
S2	**auth-url, regex rewrites, canary, modsecurity**	days per annotation type
S1	**ssl-redirect, body size, timeouts, CORS**	hours per annotation type
S0	**Vanilla Ingress - host, path, TLS only**	minutes per Ingress, automatable

The NGINX Ingress Controller ships `115+` annotations; most teams use 10–30. Audit first - 70%+ of "unsupported" ones are usually safe to drop.

09 - PLAYBOOK

The Universal Migration Pattern

01	**Map first**	List every NGINX annotation by frequency, find all snippet usage, categorise S0-S3. The estimate comes from this, nothing else.
02	**Deploy side-by-side**	New controller on its own `IngressClass`, no production traffic. Set `ingressClassName` explicitly on every Ingress.
03	**Canary DNS shift**	Move DNS or the cloud LoadBalancer to the new controller. Weighted records if your provider supports them; lower the TTL a day ahead.
04	**Bake, then clean up**	Keep the NGINX Ingress Controller running 24-48h after the DNS switch for stale resolvers. Delete the webhook first. Annotate the IngressClass `helm.sh/resource-policy: keep` or Helm takes it - and every Ingress referencing it - with the release.

LIVE DEMO

Live Migration to Traefik on AKS

An AKS cluster running the NGINX Ingress Controller: audit the annotations, deploy side-by-side, put a catch-all in front, then peel routes off one at a time.

01 - map

02 - side-by-side

03 - catch-all swing

04 - Gateway API

github.com/avivkabesa/cncf-nginx-migration-demo

10 - TAKEAWAY

What to Do Tuesday Morning

01 - Run the audit. Know what you depend on.

```
kubectl get ingress -A -o json | jq -r '
  .items[].metadata.annotations // {}
  | to_entries[]
  | select(.key | startswith(
      "nginx.ingress.kubernetes.io/"))
  | .key' | sort | uniq -c | sort -rn
```

02 - Assess your exposure

Internet-facing? Secrets in reach? Webhook exposed? Yes to all three: move now. Otherwise plan it properly.

03 - Pick your path

Already on Cilium or Istio: use what you have. Heavy annotations: Traefik. Greenfield: Envoy Gateway.

Then migrate one Ingress

The simplest, lowest-risk route. Work up the pyramid from there - do not boil the ocean.

TAKEAWAYS

01 The repository is archived and the patch path is gone - but urgency is a function of your exposure, not the date.

02 Your annotations, not your Ingress count, decide whether this is an afternoon or a quarter.

03 Swap the controller now, adopt Gateway API on your own schedule.

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Thank you

Everything from the demo is in the repo - Helm values, the catch-all route, and the Gateway API manifests.

Aviv Kabesa

demo repo
github.com/avivka

LinkedIn
in/aviv-kabesa